

B2 Symmetry and Relativity, 2025 — Model Solutions

Question 1 — Relativistic dynamics in a uniform electric field

Problem statement. List six objects; identify those that may move faster than c . Derive $\mathbf{f}(\mathbf{v}, \mathbf{a})$ for constant rest mass. A charge q , mass m , is launched from the origin with momentum $p_0 \hat{\mathbf{x}}$ in field $E_0 \hat{\mathbf{y}}$: find $\gamma(t)$, $y(t)$, $a_x(t)$, and the trajectory in the initial rest frame.

(a) Which objects can move faster than c ?

Only *signal-carrying* motion is bounded by c ; geometric or non-information-carrying features are not.

1. Neutrino: massive particle, $v < c$.
2. de Broglie wavefronts: phase velocity $c^2/v > c$, no information.
3. Scissor intersection: geometric point, no bound.
4. Cosmic ray: real particle, $v < c$.
5. Shadow edge on a screen: no signal along screen.
6. Interference fringes: stationary pattern, no information transport.

Faster-than- c allowed: 2, 3, 5, 6. Not allowed: 1, 4.

(1)

(b) Relating $\mathbf{f}$, $\mathbf{v}$ and $\mathbf{a}$

Newton's second law $\mathbf{f} = d\mathbf{p}/dt$ with $\mathbf{p} = \gamma m \mathbf{v}$ and m constant:

$$\mathbf{f} = m \frac{d(\gamma \mathbf{v})}{dt}.$$

Apply the product rule:

$$\mathbf{f} = m\gamma \mathbf{a} + m\dot{\gamma} \mathbf{v}. \tag{2}$$

Differentiate $\gamma^2 = (1 - v^2/c^2)^{-1}$ in t :

$$2\gamma\dot{\gamma} = -(1 - v^2/c^2)^{-2} \cdot (-2\mathbf{v} \cdot \mathbf{a}/c^2).$$

Simplify:

$$2\gamma\dot{\gamma} = \frac{2(\mathbf{v} \cdot \mathbf{a})/c^2}{(1 - v^2/c^2)^2}.$$

Identify $(1 - v^2/c^2)^{-2} = \gamma^4$:

$$2\gamma\dot{\gamma} = \frac{2\gamma^4(\mathbf{v} \cdot \mathbf{a})}{c^2}.$$

Divide by 2γ :

$$\dot{\gamma} = \frac{\gamma^3(\mathbf{v} \cdot \mathbf{a})}{c^2}.$$

Substitute into (??):

$$\boxed{\mathbf{f} = \gamma m \mathbf{a} + \frac{\gamma^3 m}{c^2} (\mathbf{v} \cdot \mathbf{a}) \mathbf{v}, \quad \lambda(v) = \gamma, \quad \mu(v) = \gamma^3/c^2.} \quad (3)$$

$\mathbf{a} \parallel \mathbf{f}$ iff $\mathbf{v} \cdot \mathbf{a} = 0$ (transverse) or $\mathbf{v} \parallel \mathbf{a}$ (longitudinal).

(c) The Lorentz factor: $\gamma^2 = 1 + \alpha^2$

Newton's law $d\mathbf{p}/dt = qE_0\hat{\mathbf{y}}$ with $\mathbf{p}(0) = p_0\hat{\mathbf{x}}$. Integrate:

$$\mathbf{p}(t) = p_0\hat{\mathbf{x}} + qE_0t\hat{\mathbf{y}}.$$

Use $E^2 = (mc^2)^2 + (pc)^2$ with $E = \gamma mc^2$:

$$\gamma^2 m^2 c^4 = m^2 c^4 + p^2 c^2.$$

Divide by $m^2 c^4$:

$$\gamma^2 = 1 + \left(\frac{p}{mc}\right)^2.$$

Substitute $|\mathbf{p}|^2 = p_0^2 + (qE_0t)^2$:

$$\gamma^2 = 1 + \left(\frac{p_0}{mc}\right)^2 + \left(\frac{qE_0t}{mc}\right)^2.$$

Identify α :

$$\gamma^2 = 1 + \alpha^2, \quad \boxed{\alpha = \frac{p_0}{mc}\hat{\mathbf{x}} + \frac{qE_0t}{mc}\hat{\mathbf{y}}}, \quad \gamma_0 = \sqrt{1 + (p_0/mc)^2}. \quad (4)$$

(d) Energy method: relating γ to height y , and the time of flight

Work–energy theorem with constant force $qE_0\hat{\mathbf{y}}$:

$$d(\gamma mc^2)/dt = \mathbf{f} \cdot \mathbf{v} = qE_0 dy/dt.$$

Integrate from $t = 0$ to t :

$$\gamma mc^2 - \gamma_0 mc^2 = qE_0 y.$$

Divide by mc^2 :

$$\gamma = \gamma_0 + \frac{qE_0}{mc^2} y. \quad (5)$$

Square both sides:

$$\gamma^2 = \gamma_0^2 + \frac{2\gamma_0 qE_0}{mc^2} y + \left(\frac{qE_0 y}{mc^2}\right)^2.$$

From (??), also

$$\gamma^2 = \gamma_0^2 + \left(\frac{qE_0 t}{mc}\right)^2.$$

Subtract γ_0^2 and equate:

$$\left(\frac{qE_0 t}{mc}\right)^2 = \frac{2\gamma_0 qE_0}{mc^2} y + \left(\frac{qE_0 y}{mc^2}\right)^2.$$

Multiply through by $(mc^2/qE_0)^2$:

$$c^2 t^2 = \frac{2\gamma_0 mc^2}{qE_0} y + y^2.$$

Take the square root:

$$ct = \sqrt{y^2 + \eta y}, \quad \boxed{\eta = \frac{2\gamma_0 mc^2}{qE_0}}. \quad (6)$$

Limit $y \ll \eta$: $y \approx (qE_0/2\gamma_0 m)t^2$ (effective mass $\gamma_0 m$). Limit $y \gg \eta$: $ct \rightarrow y$.

(e) Longitudinal acceleration $a_x(t)$

$p_x = \gamma m v_x = p_0$ is conserved, so

$$v_x = \frac{p_0}{\gamma m}.$$

Differentiate using $d(1/\gamma)/dt = -\dot{\gamma}/\gamma^2$:

$$a_x = \frac{dv_x}{dt} = \frac{p_0}{m} \frac{d}{dt} \left(\frac{1}{\gamma} \right).$$

Substitute the chain rule:

$$a_x = -\frac{p_0}{m} \frac{\dot{\gamma}}{\gamma^2}.$$

Differentiate $\gamma^2 = \gamma_0^2 + (qE_0 t/mc)^2$ in t :

$$2\gamma\dot{\gamma} = \frac{2(qE_0)^2 t}{m^2 c^2}.$$

Divide by 2γ :

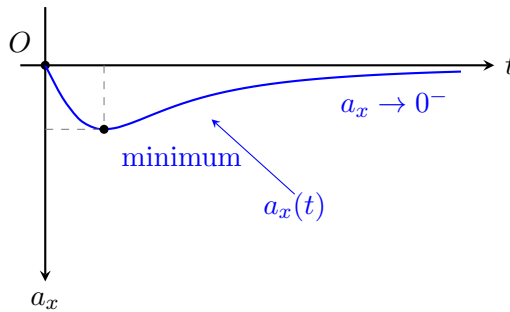
$$\dot{\gamma} = \frac{(qE_0)^2 t}{\gamma m^2 c^2}.$$

Substitute into the expression for a_x :

$$\boxed{a_x(t) = -\frac{p_0(qE_0)^2 t}{m^3 c^2 \gamma^3}, \quad \gamma^2 = \gamma_0^2 + \left(\frac{qE_0 t}{mc} \right)^2}. \quad (7)$$

At $t = 0$, $a_x = 0$; at small t , $a_x \propto -t$; at large t , $\gamma \sim qE_0 t/(mc)$ so $a_x \sim -1/t^2 \rightarrow 0^-$. Single negative extremum.

Sketch of $a_x(t)$ for $t \geq 0$:



(f) Fields and trajectory in the initial rest frame

The initial rest frame S^* moves with $\mathbf{V} = (p_0/\gamma_0 m)\hat{\mathbf{x}}$, $\gamma_V = \gamma_0$. Apply the boost formula $E_y^* = \gamma_V(E_y - VB_z)$ to $\mathbf{E} = E_0\hat{\mathbf{y}}$, $\mathbf{B} = 0$:

$$E_y^* = \gamma_0 E_0.$$

Apply $B_z^* = \gamma_V(B_z - VE_y/c^2)$:

$$B_z^* = -\gamma_0 \frac{VE_0}{c^2}.$$

Substitute $V = p_0/(\gamma_0 m)$:

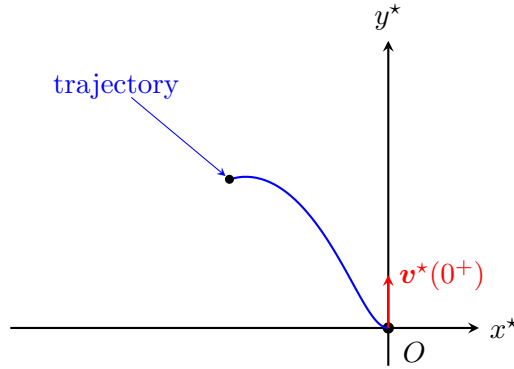
$$B_z^* = -\gamma_0 \frac{p_0}{\gamma_0 m} \frac{E_0}{c^2} = -\frac{p_0 E_0}{mc^2}.$$

Hence

$$\boxed{\mathbf{E}^* = \gamma_0 E_0 \hat{\mathbf{y}}, \quad \mathbf{B}^* = -\frac{p_0 E_0}{mc^2} \hat{\mathbf{z}}.} \quad (8)$$

At $t^* = 0$ the particle is at rest, so it accelerates along $+\hat{\mathbf{y}}$; once $v_y^* > 0$, the term $q\mathbf{v} \times \mathbf{B}^*$ deflects it towards $-\hat{\mathbf{x}}$. Trajectory enters the second quadrant.

Sketch of the spatial trajectory in S^ :*



At large t in S , $v_x = p_0/(\gamma m) \rightarrow 0$ and $v_y \rightarrow c$. Apply the inverse velocity transformation $v_x^* = (v_x - V)/(1 - v_x V/c^2)$:

$$v_x^* \rightarrow (0 - V)/(1 - 0) = -V.$$

Apply $v_y^* = v_y/[\gamma_0(1 - v_x V/c^2)]$:

$$v_y^* \rightarrow c/\gamma_0.$$

Both limits are finite, so the trajectory tends to a straight asymptotic line with slope

$$\frac{dy^*}{dx^*} = \frac{v_y^*}{v_x^*} = \frac{c/\gamma_0}{-V} = -\frac{c}{\gamma_0 V}.$$

Substitute $V = p_0/(\gamma_0 m)$:

$$\frac{dy^*}{dx^*} = -\frac{mc}{p_0}.$$

The slope is finite and nonzero, so the asymptote is straight but never vertical (which would require $v_x^* \rightarrow 0$).

Question 2 — Doppler effect and resonant absorption with crossed source/detector motion

Problem statement. Derive the relativistic Doppler formula. Then: source on $\bar{y}$ -axis at speed u through $\bar{y}_0$; detector on $\bar{x}$ -axis at speed v . Work in the detector rest frame S : find the source's worldline, v_{rel} , the emission/reception geometry, the frequency when the source crosses S 's x -axis, and the u giving resonant absorption.

(a) The Doppler formula

Take the photon 4-wavevector $K^\mu = (\omega/c)(1, \hat{\mathbf{n}})$ and the source 4-velocity $U_{\text{src}}^\mu = \gamma_\beta(c, \mathbf{v}_{\text{src}})$. The Lorentz invariant $-K \cdot U_{\text{src}}$ reduces to ω_0 in the source rest frame (where $\mathbf{v}_{\text{src}} = \mathbf{0}$). Compute the invariant in S :

$$-K \cdot U_{\text{src}} = -g_{\mu\nu} K^\mu U_{\text{src}}^\nu = K^0 U^0 - \mathbf{K} \cdot \mathbf{U}.$$

Substitute components:

$$-K \cdot U_{\text{src}} = (\omega/c)\gamma_\beta c - \gamma_\beta(\mathbf{v}_{\text{src}} \cdot \hat{\mathbf{n}})(\omega/c).$$

Substitute $\mathbf{v}_{\text{src}} \cdot \hat{\mathbf{n}} = v_{\text{src}} \cos \theta$ and factor:

$$-K \cdot U_{\text{src}} = \gamma_\beta \omega (1 - \beta \cos \theta).$$

Equate to ω_0 :

$$\omega_0 = \gamma_\beta \omega (1 - \beta \cos \theta).$$

Solve for ω using $\gamma_\beta = 1/\sqrt{1 - \beta^2}$:

$$\boxed{\omega = \frac{\omega_0 \sqrt{1 - \beta^2}}{1 - \beta \cos \theta}}. \quad (9)$$

(b) Trajectory of the source in the detector frame S

The detector at the origin of S moves along $+\hat{x}$ at speed v in $\bar{S}$, so S moves at $+v\hat{x}$ relative to $\bar{S}$ with $\gamma = (1 - v^2/c^2)^{-1/2}$. Source worldline in $\bar{S}$: $\bar{x} = 0$, $\bar{y} = \bar{y}_0 + u\bar{t}$, $\bar{z} = 0$. Apply the Lorentz transformation $\bar{S} \rightarrow S$ at $\bar{x} = 0$:

$$t = \gamma(\bar{t} - v\bar{x}/c^2) = \gamma\bar{t}.$$

Apply the transformation for x :

$$x = \gamma(\bar{x} - v\bar{t}) = -\gamma v\bar{t}.$$

Substitute $\bar{t} = t/\gamma$:

$$x = -v t.$$

Apply $y = \bar{y}$ (transverse coordinate unchanged):

$$y = \bar{y}_0 + u\bar{t} = \bar{y}_0 + u t/\gamma.$$

Check $t = 0$ gives $\bar{t} = 0$, hence $(x, y, z) = (0, \bar{y}_0, 0)$. Combine:

$$\boxed{\mathbf{r}_{\text{src}}(t) = (-vt, \bar{y}_0 + ut/\gamma, 0)}. \quad (10)$$

(c) Velocity of the source in S and relative speed

Differentiate (??) in t :

$$\mathbf{v}_{\text{src}}^{(S)} = (-v, u/\gamma, 0).$$

The detector is at rest in S , so the relative speed equals $|\mathbf{v}_{\text{src}}^{(S)}|$:

$$v_{\text{rel}}^2 = v^2 + (u/\gamma)^2.$$

Substitute $1/\gamma^2 = 1 - v^2/c^2$:

$$v_{\text{rel}}^2 = v^2 + u^2(1 - v^2/c^2).$$

Take the square root:

$$v_{\text{rel}} = \sqrt{v^2 + u^2(1 - v^2/c^2)}. \quad (11)$$

Limits: $v, u \ll c \Rightarrow \sqrt{v^2 + u^2}$; $u \rightarrow c \Rightarrow v_{\text{rel}} \rightarrow c$.

(d) Emission/reception geometry

A photon emitted at the source position $\mathbf{r}_{\text{src}}(t)$ reaches the detector at the origin at time t_r with light travel condition $|\mathbf{r}_{\text{src}}(t)| = c(t_r - t)$. Substitute components from (??):

$$c(t_r - t) = \sqrt{v^2 t^2 + (\bar{y}_0 + ut/\gamma)^2}. \quad (12)$$

The source's velocity in S is along its trajectory:

$$\mathbf{V}_{\text{src}} = (-v, u/\gamma, 0).$$

The light ray from source to detector is along $-\mathbf{r}_{\text{src}}(t)$:

$$-\mathbf{r}_{\text{src}}(t) = (vt, -\bar{y}_0 - ut/\gamma, 0), \quad |-\mathbf{r}_{\text{src}}| = c(t_r - t).$$

Let θ be the angle between $\mathbf{V}_{\text{src}}$ and $-\mathbf{r}_{\text{src}}$:

$$v_{\text{rel}} c(t_r - t) \cos \theta = \mathbf{V}_{\text{src}} \cdot (-\mathbf{r}_{\text{src}}(t)).$$

Expand the dot product:

$$\mathbf{V}_{\text{src}} \cdot (-\mathbf{r}_{\text{src}}) = (-v)(vt) + (u/\gamma)(-\bar{y}_0 - ut/\gamma).$$

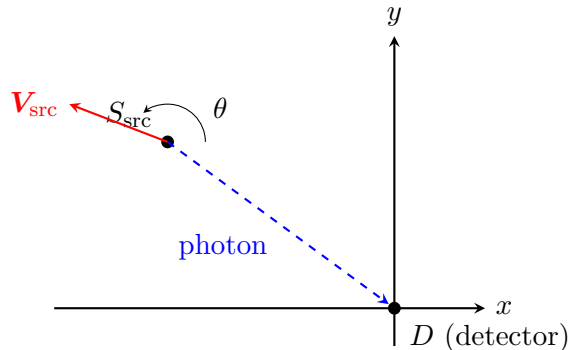
Distribute the products:

$$\mathbf{V}_{\text{src}} \cdot (-\mathbf{r}_{\text{src}}) = -v^2 t - \frac{u\bar{y}_0}{\gamma} - \frac{u^2 t}{\gamma^2}.$$

Collect:

$$v_{\text{rel}} c(t_r - t) \cos \theta = -v^2 t - \frac{u\bar{y}_0}{\gamma} - \frac{u^2 t}{\gamma^2}. \quad (13)$$

Geometry in the detector frame S :



(e) Frequency observed when the source crosses the x -axis

Set $y(t_\star) = 0$ in (??):

$$\bar{y}_0 + ut_\star/\gamma = 0.$$

Solve for $t_\star$:

$$t_\star = -\gamma\bar{y}_0/u.$$

At this instant $\mathbf{r}_{\text{src}}(t_\star) = (-vt_\star, 0, 0)$. Taking $\bar{y}_0 > 0$, $u > 0$ gives $t_\star < 0$, so the source is at $x_\star = -vt_\star > 0$ and the source-to-detector light ray propagates along $-\hat{\mathbf{x}}$. Project the source velocity onto this direction:

$$\mathbf{V}_{\text{src}} \cdot (-\hat{\mathbf{x}}) = (-v, u/\gamma, 0) \cdot (-1, 0, 0) = v.$$

Use $|\mathbf{V}_{\text{src}}| = v_{\text{rel}}$ to identify $\cos \theta$:

$$\cos \theta = \frac{v}{v_{\text{rel}}}.$$

Substitute $\beta = v_{\text{rel}}/c$ into (??):

$$\omega = \frac{\omega_0 \sqrt{1 - v_{\text{rel}}^2/c^2}}{1 - (v_{\text{rel}}/c) \cos \theta}.$$

Use $v_{\text{rel}} \cos \theta = v$:

$$\omega = \frac{\omega_0 \sqrt{1 - v_{\text{rel}}^2/c^2}}{1 - v/c}.$$

Expand $1 - v_{\text{rel}}^2/c^2$ using $v_{\text{rel}}^2 = v^2 + u^2(1 - v^2/c^2)$:

$$1 - v_{\text{rel}}^2/c^2 = 1 - v^2/c^2 - u^2/c^2 + u^2v^2/c^4.$$

Factor:

$$1 - v_{\text{rel}}^2/c^2 = (1 - v^2/c^2)(1 - u^2/c^2).$$

Substitute back:

$$\omega = \omega_0 \frac{\sqrt{(1 - v^2/c^2)(1 - u^2/c^2)}}{1 - v/c}.$$

Factor $1 - v^2/c^2 = (1 - v/c)(1 + v/c)$:

$$\omega = \omega_0 \sqrt{\frac{(1 - v/c)(1 + v/c)(1 - u^2/c^2)}{(1 - v/c)^2}}.$$

Cancel one factor of $(1 - v/c)$:

$$\boxed{\omega = \omega_0 \sqrt{\frac{(1 + v/c)(1 - u^2/c^2)}{1 - v/c}}.} \quad (14)$$

(f) Resonant absorption

Read the paper's specification literally: ground-state rest mass m , excited-state rest mass $m + \hbar\omega_0/c^2$ (treating $\hbar\omega_0$ as the excitation energy). Pre-collision 4-momenta in S : atom at rest $\mathbf{P}_{\text{atom}} = (mc, \mathbf{0})$, photon travelling along $-\hat{\mathbf{x}}$ with $\mathbf{P}_\gamma = (\hbar\omega/c, -\hbar\omega/c, 0, 0)$. Sum:

$$\mathbf{P}_{\text{tot}} = (mc + \hbar\omega/c, -\hbar\omega/c, 0, 0).$$

The excited atom carries this total 4-momentum, so $P_{\text{tot}}^2 = -M^2c^2$ with $M = m + \hbar\omega_0/c^2$. Compute:

$$P_{\text{tot}}^2 = -(mc + \hbar\omega/c)^2 + (\hbar\omega/c)^2 = -M^2c^2.$$

Multiply by -1 :

$$(mc + \hbar\omega/c)^2 - (\hbar\omega/c)^2 = M^2c^2.$$

Expand the first square:

$$m^2c^2 + 2m\hbar\omega + (\hbar\omega/c)^2 - (\hbar\omega/c)^2 = M^2c^2.$$

Cancel the $(\hbar\omega/c)^2$ terms:

$$m^2c^2 + 2m\hbar\omega = M^2c^2.$$

Substitute $M = m + \hbar\omega_0/c^2$ and expand M^2c^2 :

$$m^2c^2 + 2m\hbar\omega = m^2c^2 + 2m\hbar\omega_0 + \hbar^2\omega_0^2/c^2.$$

Cancel m^2c^2 :

$$2m\hbar\omega = 2m\hbar\omega_0 + \hbar^2\omega_0^2/c^2.$$

Divide by $2m\hbar$:

$$\omega = \omega_0 + \frac{\hbar\omega_0^2}{2mc^2}.$$

Factor out ω_0 :

$$\omega_{\text{res}} = \omega_0 \left(1 + \frac{\hbar\omega_0}{2mc^2} \right). \quad (15)$$

Equate (??) with (??):

$$\omega_0 \sqrt{\frac{(1 + v/c)(1 - u^2/c^2)}{1 - v/c}} = \omega_0 \left(1 + \frac{\hbar\omega_0}{2mc^2} \right).$$

Divide by ω_0 and square:

$$\frac{(1 + v/c)(1 - u^2/c^2)}{1 - v/c} = \left(1 + \frac{\hbar\omega_0}{2mc^2} \right)^2.$$

Multiply by $(1 - v/c)/(1 + v/c)$:

$$1 - u^2/c^2 = \frac{1 - v/c}{1 + v/c} \left(1 + \frac{\hbar\omega_0}{2mc^2} \right)^2.$$

Solve for u :

$$u = c \sqrt{1 - \frac{1 - v/c}{1 + v/c} \left(1 + \frac{\hbar\omega_0}{2mc^2} \right)^2}. \quad (16)$$

Question 3 — 4-vectors, EM tensor, and stress-energy

Problem statement. Identify which four-tuples are 4-vectors. Show that a 4-force preserves rest mass iff $F \cdot U = 0$, and deduce antisymmetry of $\mathbb{G}$. With $\mathbf{E} \parallel \mathbf{B}$, $|\mathbf{E}| = E_0$, $|\mathbf{B}| = B_0 = E_0/c$ in S , study the EM invariants and the stress-energy tensor in S and a boosted frame S' .

(a) Which are 4-vectors?

Multiplication of a 4-vector by a Lorentz scalar yields a 4-vector; multiplication by a frame-dependent quantity does not. For a particle, $\mathbf{p} = \gamma m \mathbf{v} = (E/c^2) \mathbf{v}$, so $E \mathbf{v} = c^2 \mathbf{p}$.

- **1.** $X^\mu = (ct, \mathbf{r})$ – yes (the position 4-vector).
- **2.** $(c, \mathbf{v})$ – no; the 4-velocity is $\gamma(c, \mathbf{v})$, and $1/\gamma$ is frame-dependent.
- **3.** $P^\mu = (E/c, \mathbf{p})$ – yes (4-momentum).
- **4.** mX^μ – yes (m is a Lorentz invariant).
- **5.** $m(c, \mathbf{v})$ – no; again missing the γ factor.
- **6.** $E(c, \mathbf{v}) = (Ec, E\mathbf{v})$. Use $E\mathbf{v} = c^2 \mathbf{p}$:

$$(Ec, E\mathbf{v}) = (cE, c^2 \mathbf{p}).$$

Factor out c^2 :

$$(Ec, E\mathbf{v}) = c^2 (E/c, \mathbf{p}) = c^2 P^\mu.$$

Since c^2 is an invariant constant, this *is* a 4-vector.

4-vectors: 1, 3, 4, 6. Not 4-vectors: 2, 5.

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(b) Rest-mass preservation

Start from $U \cdot U = -c^2$ (Lorentz invariant). Differentiate in proper time τ :

$$2 A \cdot U = 0.$$

Divide by 2:

$$A \cdot U = 0.$$

Write the 4-momentum $\mathbf{P} = mU$ and differentiate:

$$\mathbf{F} = \dot{m} U + m A.$$

Dot with U :

$$F \cdot U = \dot{m} (U \cdot U) + m (A \cdot U).$$

Substitute $U \cdot U = -c^2$ and $A \cdot U = 0$:

$$F \cdot U = -\dot{m} c^2.$$

Hence $F \cdot U = 0 \Leftrightarrow \dot{m} = 0$.

For $\mathbf{F} = q \mathbb{G} \cdot U$, write the contraction with U using the metric convention $F \cdot U = F^a g_{ab} U^b = (q \mathbb{G}^{ac} U_c) U_a$:

$$F \cdot U = q U_a U_c \mathbb{G}^{ac}.$$

Decompose $\mathbb{G}^{ac} = \mathbb{G}^{(ac)} + \mathbb{G}^{[ac]}$. The symmetric tensor $U_a U_c$ contracted with the antisymmetric part vanishes:

$$F \cdot U = q U_a U_c \mathbb{G}^{(ac)}.$$

For this to vanish for *every* 4-velocity U , the symmetric part must be zero, i.e.

$\mathbb{G}^{ab} = -\mathbb{G}^{ba}$.

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This is why the Faraday tensor is antisymmetric.

(c) **Invariants of the EM field with parallel $\mathbf{E}, \mathbf{B}$**

The two Lorentz-scalar combinations are

$$I_1 = \mathbf{B}^2 - \mathbf{E}^2/c^2, \quad I_2 = \mathbf{E} \cdot \mathbf{B}. \quad (19)$$

In S , with $\mathbf{E} \parallel \mathbf{B}$, $|\mathbf{E}| = E_0$, $|\mathbf{B}| = E_0/c$, evaluate I_1 :

$$I_1 = (E_0/c)^2 - E_0^2/c^2 = 0.$$

Evaluate I_2 :

$$I_2 = E_0 (E_0/c) = E_0^2/c.$$

Invariance of I_1 in any other frame $S^\dagger$ gives $|\mathbf{B}^\dagger|^2 = |\mathbf{E}^\dagger|^2/c^2$:

$$|\mathbf{E}^\dagger| = c|\mathbf{B}^\dagger|.$$

Invariance of I_2 gives

$$\mathbf{E}^\dagger \cdot \mathbf{B}^\dagger = E_0^2/c.$$

Apply Cauchy–Schwarz, $\mathbf{E}^\dagger \cdot \mathbf{B}^\dagger \leq |\mathbf{E}^\dagger||\mathbf{B}^\dagger|$:

$$E_0^2/c \leq |\mathbf{E}^\dagger||\mathbf{B}^\dagger|.$$

Substitute $|\mathbf{E}^\dagger| = c|\mathbf{B}^\dagger|$:

$$E_0^2/c \leq c|\mathbf{B}^\dagger|^2.$$

Solve for $|\mathbf{B}^\dagger|$:

$$|\mathbf{B}^\dagger|^2 \geq E_0^2/c^2.$$

Take the positive square root and use $|\mathbf{E}^\dagger| = c|\mathbf{B}^\dagger|$:

$$\boxed{|\mathbf{B}^\dagger| \geq B_0 = E_0/c, \quad |\mathbf{E}^\dagger| \geq E_0.}$$

Given $|\mathbf{B}'| = \sqrt{2} B_0$, use $|\mathbf{E}'| = c|\mathbf{B}'|$:

$$|\mathbf{E}'| = c\sqrt{2} B_0 = \sqrt{2} E_0.$$

Apply the definition of $\cos \phi$:

$$\cos \phi = \frac{\mathbf{E}' \cdot \mathbf{B}'}{|\mathbf{E}'||\mathbf{B}'|}.$$

Substitute $\mathbf{E}' \cdot \mathbf{B}' = I_2 = E_0^2/c$ and $|\mathbf{E}'||\mathbf{B}'| = (\sqrt{2} E_0)(\sqrt{2} E_0/c)$:

$$\cos \phi = \frac{E_0^2/c}{2E_0^2/c}.$$

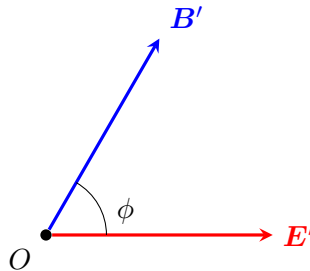
Cancel:

$$\cos \phi = \frac{1}{2}.$$

Hence

$$\boxed{\phi = \pi/3.} \quad (20)$$

Sketch of $\mathbf{E}'$ and $\mathbf{B}'$ in S' :



(d) **Stress-energy tensor in S**

In S , $\mathbf{E} = E_0 \hat{\mathbf{y}}$, $\mathbf{B} = (E_0/c) \hat{\mathbf{y}}$. The non-zero components of $F^{\mu\nu}$ are

$$F^{02} = E_0/c, \quad F^{20} = -E_0/c, \quad F^{13} = -B_y = -E_0/c, \quad F^{31} = E_0/c.$$

Compute the invariant $F^{\mu\nu} F_{\mu\nu}$ using $F_{0i} = -F^{0i}$, $F_{ij} = F^{ij}$:

$$F^{\mu\nu} F_{\mu\nu} = 2F^{02} F_{02} + 2F^{13} F_{13} = -2(E_0/c)^2 + 2(E_0/c)^2.$$

Collect:

$$F^{\mu\nu} F_{\mu\nu} = 0 = 2I_1.$$

Hence the trace-piece $-\frac{1}{4}g^{ab}F^{\mu\nu}F_{\mu\nu}$ vanishes, leaving $\mathbb{T}^{ab} = -\epsilon_0 c^2 g^{a\lambda} F_{\lambda\mu} F^{\mu b}$.

Energy density $\mathbb{T}^{00} = \epsilon_0 c^2 F_{0\mu} F^{\mu 0}$:

$$\mathbb{T}^{00} = \epsilon_0 c^2 F_{02} F^{20}.$$

Substitute $F_{02} = -E_0/c$, $F^{20} = -E_0/c$:

$$\mathbb{T}^{00} = \epsilon_0 c^2 \cdot E_0^2/c^2 = \epsilon_0 E_0^2.$$

This equals the field energy density $\frac{1}{2}\epsilon_0 E_0^2 + B_0^2/(2\mu_0) = \epsilon_0 E_0^2$. For the energy-flux/momentum-density entries $\mathbb{T}^{0i} = \epsilon_0 c^2 F_{0\mu} F^{\mu i}$, the only nonzero $F_{0\mu}$ component is F_{02} , and F^{2i} vanishes for every spatial i (since $B_x = B_z = 0$ and E_y does not enter F^{2i}):

$$\mathbb{T}^{0i} = 0 \quad (\text{Poynting vector } \mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0 = 0).$$

For $\mathbb{T}^{xx} = -\epsilon_0 c^2 F_{1\mu} F^{\mu 1}$, only $\mu = 3$ contributes:

$$\mathbb{T}^{xx} = -\epsilon_0 c^2 F_{13} F^{31}.$$

Substitute $F_{13} = -E_0/c$, $F^{31} = E_0/c$:

$$\mathbb{T}^{xx} = -\epsilon_0 c^2 \cdot (-E_0^2/c^2) = +\epsilon_0 E_0^2.$$

For $\mathbb{T}^{yy} = -\epsilon_0 c^2 F_{2\mu} F^{\mu 2}$, only $\mu = 0$ contributes:

$$\mathbb{T}^{yy} = -\epsilon_0 c^2 F_{20} F^{02} = -\epsilon_0 c^2 \cdot E_0^2/c^2 = -\epsilon_0 E_0^2.$$

By the same calculation as $\mathbb{T}^{xx}$ (with $F_{31} F^{13}$):

$$\mathbb{T}^{zz} = +\epsilon_0 E_0^2.$$

Assemble:

$$\mathbb{T} = \epsilon_0 E_0^2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (21)$$

$$\boxed{\mathbb{T}^{00} = \epsilon_0 E_0^2, \quad \mathbb{T}^{0i} = 0, \quad \mathbb{T}^{xx} = \mathbb{T}^{zz} = +\epsilon_0 E_0^2, \quad \mathbb{T}^{yy} = -\epsilon_0 E_0^2.} \quad (22)$$

Interpretation: $\mathbb{T}^{00}$ is energy density; $c\mathbb{T}^{0i}$ is energy flux (zero, since parallel $\mathbf{E}, \mathbf{B}$); $\mathbb{T}^{ij}$ is momentum-flux/stress. $\mathbb{T}^{yy} < 0$ encodes tension along the field lines; $\mathbb{T}^{xx}, \mathbb{T}^{zz} > 0$ encode pressure transverse to them. Symmetry $\mathbb{T}^{0i} = \mathbb{T}^{i0}$: momentum density equals energy flux divided by c^2 .

(e) Stress-energy in S'

Boost along $\hat{\mathbf{x}}$ at $\beta = v/c$. The Lorentz matrix has nonzero entries

$$\Lambda^0_0 = \Lambda^1_1 = \gamma, \quad \Lambda^0_1 = \Lambda^1_0 = -\gamma\beta,$$

and $\Lambda^2_2 = \Lambda^3_3 = 1$. Apply $\mathbb{T}'^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta \mathbb{T}^{\alpha\beta}$ to the (t, x) -block. Compute $\mathbb{T}'^{00}$:

$$\mathbb{T}'^{00} = (\Lambda^0_0)^2 \mathbb{T}^{00} + (\Lambda^0_1)^2 \mathbb{T}^{xx}.$$

Substitute:

$$\mathbb{T}'^{00} = \gamma^2 \mathbb{T}^{00} + \gamma^2 \beta^2 \mathbb{T}^{xx} = \epsilon_0 E_0^2 \gamma^2 (1 + \beta^2).$$

Compute $\mathbb{T}'^{0x}$ (cross terms via symmetry of $\mathbb{T}$):

$$\mathbb{T}'^{0x} = \Lambda^0_0 \Lambda^1_0 \mathbb{T}^{00} + \Lambda^0_1 \Lambda^1_1 \mathbb{T}^{xx}.$$

Substitute:

$$\mathbb{T}'^{0x} = -\gamma^2 \beta \mathbb{T}^{00} - \gamma^2 \beta \mathbb{T}^{xx} = -\gamma^2 \beta (\mathbb{T}^{00} + \mathbb{T}^{xx}).$$

Insert $\mathbb{T}^{00} = \mathbb{T}^{xx} = \epsilon_0 E_0^2$:

$$\mathbb{T}'^{0x} = -2\epsilon_0 E_0^2 \gamma^2 \beta.$$

Compute $\mathbb{T}'^{xx}$:

$$\mathbb{T}'^{xx} = (\Lambda^1_0)^2 \mathbb{T}^{00} + (\Lambda^1_1)^2 \mathbb{T}^{xx} = \gamma^2 \beta^2 \mathbb{T}^{00} + \gamma^2 \mathbb{T}^{xx}.$$

Simplify:

$$\mathbb{T}'^{xx} = \epsilon_0 E_0^2 \gamma^2 (1 + \beta^2).$$

The y - and z -blocks are unaffected by the x -boost (their Λ rows act trivially), so $\mathbb{T}'^{yy} = -\epsilon_0 E_0^2$, $\mathbb{T}'^{zz} = +\epsilon_0 E_0^2$, and all remaining off-diagonals are zero. Hence

$$\boxed{\mathbb{T}' = \epsilon_0 E_0^2 \begin{pmatrix} \gamma^2(1 + \beta^2) & -2\gamma^2\beta & 0 & 0 \\ -2\gamma^2\beta & \gamma^2(1 + \beta^2) & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}. \quad (23)$$

(f) Energy flux at low velocity

To first order in v/c , take $\gamma \approx 1$, so $\gamma^2 \beta \approx v/c$. Read off the energy flux cT'^{0x} :

$$cT'^{0x} = -2c \gamma^2 \beta \epsilon_0 E_0^2 \approx -2v \epsilon_0 E_0^2.$$

Energy density unchanged at this order:

$$T'^{00} \approx \epsilon_0 E_0^2 \equiv \rho.$$

In S' the field configuration drifts at velocity $-v\hat{\mathbf{x}}$. The convection piece $\rho \mathbf{v}_{\text{field}}$ contributes

$$\rho \cdot (-v) = -\epsilon_0 E_0^2 v,$$

which is only half of cT'^{0x} . The remainder comes from the work done by the Maxwell stress: the spatial stress tensor in S has $T^{xx} = +\epsilon_0 E_0^2$ (lateral pressure transverse to $\mathbf{E}$). A surface moving with velocity $-v\hat{\mathbf{x}}$ against a pressure T^{xx} transports energy at rate $T^{xx} \cdot (-v)$:

$$T^{xx} \cdot (-v) = -\epsilon_0 E_0^2 v.$$

Add the two contributions:

$$\boxed{cT'^{0x} = \underbrace{\rho \cdot (-v)}_{-\epsilon_0 E_0^2 v} + \underbrace{T^{xx} \cdot (-v)}_{-\epsilon_0 E_0^2 v} = -2\epsilon_0 E_0^2 v}. \quad (24)$$

Question 4 — Mandelstam variables, scattering kinematics, and free-particle action

Problem statement. Elastic $P + Q \rightarrow P' + Q'$ with rest masses m_1, m_2 . Develop Mandelstam variables, prove ZMF energy conservation, find $t, s, \cos \theta$. With $c = 1$, treat the relativistic free particle: maximal-proper-time worldline, Lagrangian, Hamiltonian.

(a) Lorentz-invariant scalar products

Conservation: $P + Q = P' + Q'$. Square both sides:

$$(P + Q)^2 = (P' + Q')^2.$$

Expand:

$$P^2 + 2P \cdot Q + Q^2 = P'^2 + 2P' \cdot Q' + Q'^2.$$

Substitute mass shells $P^2 = P'^2 = -m_1^2 c^2$, $Q^2 = Q'^2 = -m_2^2 c^2$:

$$-m_1^2 c^2 + 2P \cdot Q - m_2^2 c^2 = -m_1^2 c^2 + 2P' \cdot Q' - m_2^2 c^2.$$

Cancel mass-shell terms and divide by 2:

$$\boxed{P \cdot Q = P' \cdot Q'}. \quad (25)$$

Rewrite conservation as $P - Q' = P' - Q$. Square:

$$(P - Q')^2 = (P' - Q)^2.$$

Expand:

$$P^2 - 2P \cdot Q' + Q'^2 = P'^2 - 2P' \cdot Q + Q^2.$$

Substitute mass shells and cancel:

$$\boxed{P \cdot Q' = P' \cdot Q}. \quad (26)$$

(b) The sum $s + t + u$

With $c = 1$ and signature $(-, +, +, +)$, $P^2 = -m_1^2$, $P'^2 = -m_1^2$, $Q^2 = -m_2^2$, $Q'^2 = -m_2^2$. Expand $s = -(P + Q)^2$:

$$s = -P^2 - 2P \cdot Q - Q^2.$$

Substitute mass shells:

$$s = m_1^2 + m_2^2 - 2P \cdot Q.$$

Expand $t = -(P - P')^2$:

$$t = -P^2 + 2P \cdot P' - P'^2 = 2m_1^2 + 2P \cdot P'.$$

Expand $u = -(P - Q')^2$:

$$u = -P^2 + 2P \cdot Q' - Q'^2 = m_1^2 + m_2^2 + 2P \cdot Q'.$$

Add the three:

$$s + t + u = 4m_1^2 + 2m_2^2 + 2P \cdot (-Q + P' + Q').$$

Use conservation $P' + Q' = P + Q$:

$$-Q + P' + Q' = P.$$

Substitute:

$$s + t + u = 4m_1^2 + 2m_2^2 + 2\mathbf{P} \cdot \mathbf{P}.$$

Substitute $\mathbf{P} \cdot \mathbf{P} = -m_1^2$:

$$s + t + u = 4m_1^2 + 2m_2^2 - 2m_1^2.$$

Collect:

$$\boxed{s + t + u = 2m_1^2 + 2m_2^2.} \quad (27)$$

(c) Energy conservation in the zero-momentum frame

In the ZMF, $\mathbf{p} + \mathbf{q} = 0$ gives $\mathbf{q} = -\mathbf{p}$, and similarly $\mathbf{q}' = -\mathbf{p}'$. Hence $|\mathbf{q}| = |\mathbf{p}| \equiv p$ and $|\mathbf{q}'| = |\mathbf{p}'| \equiv p'$. Use $E_i^2 = m_i^2 + |\mathbf{p}_i|^2$:

$$E_1^2 - E_2^2 = (m_1^2 + p^2) - (m_2^2 + p^2).$$

Cancel p^2 :

$$E_1^2 - E_2^2 = m_1^2 - m_2^2.$$

Similarly for the outgoing pair:

$$E_1'^2 - E_2'^2 = m_1^2 - m_2^2.$$

Equate:

$$E_1^2 - E_2^2 = E_1'^2 - E_2'^2.$$

Factor as a difference of squares:

$$(E_1 - E_2)(E_1 + E_2) = (E_1' - E_2')(E_1' + E_2').$$

Apply total-energy conservation $E_1 + E_2 = E_1' + E_2'$ and divide by this common factor:

$$E_1 - E_2 = E_1' - E_2'.$$

Add to $E_1 + E_2 = E_1' + E_2'$ and divide by 2:

$$E_1 = E_1'.$$

Subtract and divide by 2:

$$E_2 = E_2'.$$

Conclude:

$$\boxed{E_1 = E_1', \quad E_2 = E_2',} \quad (28)$$

and consequently $|\mathbf{p}| = |\mathbf{p}'|$, i.e. $p = p'$.

(d) t and $\cos \theta$; s in terms of energies

In the ZMF, part (c) gives $E_1 = E_1'$, so the 4-vector difference reduces to its spatial part:

$$\mathbf{P} - \mathbf{P}' = (0, \mathbf{p} - \mathbf{p}').$$

Note also $\mathbf{q}' - \mathbf{q} = (-\mathbf{p}') - (-\mathbf{p}) = \mathbf{p} - \mathbf{p}'$, so the angle between $\mathbf{q}, \mathbf{q}'$ equals that between $\mathbf{p}, \mathbf{p}'$. Square (signature $(-, +, +, +)$):

$$(\mathbf{P} - \mathbf{P}')^2 = |\mathbf{p} - \mathbf{p}'|^2.$$

Expand the square using $|\mathbf{p}| = |\mathbf{p}'| = p$ and $\mathbf{p} \cdot \mathbf{p}' = p^2 \cos \theta$:

$$|\mathbf{p} - \mathbf{p}'|^2 = p^2 - 2p^2 \cos \theta + p^2.$$

Collect:

$$|\mathbf{p} - \mathbf{p}'|^2 = 2p^2(1 - \cos \theta).$$

Apply $t = -(\mathbf{P} - \mathbf{P}')^2$:

$$t = -2p^2(1 - \cos \theta). \quad (29)$$

In the ZMF the total 4-momentum is $(E_1 + E_2, \mathbf{0})$:

$$s = -(\mathbf{P} + \mathbf{Q})^2 = -[-(E_1 + E_2)^2] = (E_1 + E_2)^2.$$

Expand the square:

$$s = E_1^2 + 2E_1E_2 + E_2^2.$$

Substitute $E_i^2 = m_i^2 + p^2$:

$$s = m_1^2 + m_2^2 + 2E_1E_2 + 2p^2.$$

This proves the desired identity. From $E_1 + E_2 = \sqrt{s}$ and $E_1 - E_2 = (m_1^2 - m_2^2)/\sqrt{s}$ (part (c)):

$$E_1 = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}.$$

And

$$E_2 = \frac{s - m_1^2 + m_2^2}{2\sqrt{s}}.$$

Multiply (difference of squares):

$$E_1E_2 = \frac{s^2 - (m_1^2 - m_2^2)^2}{4s}.$$

Rearrange $s = m_1^2 + m_2^2 + 2E_1E_2 + 2p^2$ for $2p^2$:

$$2p^2 = s - m_1^2 - m_2^2 - 2E_1E_2.$$

Substitute $2E_1E_2$:

$$2p^2 = s - m_1^2 - m_2^2 - \frac{s^2 - (m_1^2 - m_2^2)^2}{2s}.$$

Multiply through by $2s$:

$$4sp^2 = 2s(s - m_1^2 - m_2^2) - s^2 + (m_1^2 - m_2^2)^2.$$

Expand and collect:

$$4sp^2 = s^2 - 2s(m_1^2 + m_2^2) + (m_1^2 - m_2^2)^2.$$

Factor as the Källén function $\lambda(s, m_1^2, m_2^2) = [s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]$:

$$p^2 = \frac{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}{4s}.$$

Solve (??) for $1 - \cos \theta$:

$$1 - \cos \theta = -\frac{t}{2p^2}.$$

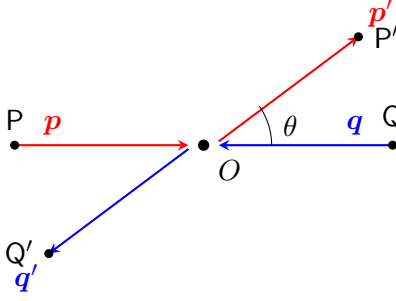
Substitute p^2 :

$$1 - \cos \theta = -\frac{2st}{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}.$$

Rearrange:

$$\boxed{\cos \theta = 1 + \frac{2st}{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}}. \quad (30)$$

ZMF scattering geometry:



(e) Application to $p\bar{p} \rightarrow \pi\bar{\pi}$ and Lorentz invariance of probabilities

Masses change ($m_p \neq m_\pi$), so part (a) no longer gives $E_1 = E'_1$. The Mandelstam variables are still Lorentz invariants with $s + t + u = 2m_p^2 + 2m_\pi^2$, and an analogous formula holds for $\cos \theta(s, t, m_p, m_\pi)$; $|\mathbf{p}|$ differs for in/out pairs.

A probability of obtaining $\pi\bar{\pi}$ versus $p\bar{p}$ is the ratio of two Lorentz-scalar quantities ($|\mathcal{M}|^2$ integrated over invariant phase space, divided by an invariant flux), so it is itself frame-invariant. Equivalently, “which final state occurred” is a frame-independent fact.

(f) Worldline of maximum proper time

The proper time along a timelike worldline from event A to event B :

$$\tau_{AB} = \int_A^B \frac{\sqrt{-dX \cdot dX}}{c} = \int_A^B \sqrt{1 - v^2/c^2} dt.$$

The integrand $\sqrt{1 - v^2/c^2} \leq 1$ is maximised pointwise when $v = 0$ in the chosen frame. By the inverse triangle inequality for timelike vectors (any kink or curve shortens proper time), the global maximum is achieved by the unique unaccelerated worldline:

$$\boxed{\text{Straight (inertial) worldline through } A \text{ and } B.} \quad (31)$$

(g) Lagrangian and Hamiltonian for the free relativistic particle

Take (with $c = 1$)

$$L = -m\sqrt{1 - \mathbf{v}^2}. \quad (32)$$

Since $d\tau = \sqrt{1 - \mathbf{v}^2} dt$, the action becomes

$$S = \int_{t_1}^{t_2} L dt = -m \int_{t_1}^{t_2} \sqrt{1 - \mathbf{v}^2} dt = -m \int d\tau.$$

Stationary S corresponds to extremal $\int d\tau$, i.e. the maximal-proper-time (inertial) worldline from part (f). Compute the canonical momentum component-wise:

$$p_i = \frac{\partial L}{\partial v_i} = -m \frac{\partial}{\partial v_i} \sqrt{1 - \mathbf{v}^2}.$$

Differentiate:

$$p_i = -m \cdot \frac{-v_i}{\sqrt{1 - \mathbf{v}^2}} = \frac{mv_i}{\sqrt{1 - \mathbf{v}^2}}.$$

Collect into vector form:

$$\mathbf{p} = \frac{m\mathbf{v}}{\sqrt{1 - \mathbf{v}^2}} = \gamma m\mathbf{v}.$$

Apply the Legendre transform $H = \mathbf{p} \cdot \mathbf{v} - L$:

$$H = \frac{m\mathbf{v}^2}{\sqrt{1-\mathbf{v}^2}} - (-m\sqrt{1-\mathbf{v}^2}).$$

Combine over a common denominator:

$$H = \frac{m\mathbf{v}^2 + m(1-\mathbf{v}^2)}{\sqrt{1-\mathbf{v}^2}}.$$

Cancel $\mathbf{v}^2$ in the numerator:

$$H = \frac{m}{\sqrt{1-\mathbf{v}^2}} = \gamma m.$$

Express in terms of canonical variables. From $\mathbf{p} = \gamma m\mathbf{v}$ and $\gamma = 1/\sqrt{1-\mathbf{v}^2}$:

$$\gamma^2 m^2 = m^2 + \mathbf{p}^2.$$

Take the positive root:

$$\gamma m = \sqrt{m^2 + \mathbf{p}^2}.$$

Substitute:

$$\boxed{H(\mathbf{p}, \mathbf{r}) = \sqrt{\mathbf{p}^2 + m^2} \quad (c = 1), \quad \text{or} \quad H = \sqrt{\mathbf{p}^2 c^2 + m^2 c^4}.} \quad (33)$$

Non-relativistic limit: $H \approx mc^2 + \mathbf{p}^2/(2m)$.